JUSTIN I. CHEN

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EDUCATION

University of California, San Diego (UCSD) *Sep. 2023 – Jun. 2027*

Bachelor of Science in Electrical Engineering

- **GPA:** 3.94 / 4.0
- **Relevant Coursework:** C Programming, Object Oriented Programming, Digital Systems and Design, Assembly and Computer Architecture, Circuits and Analog Design, Linear Algebra, Multivariable Calculus, Diff Eq, Vector Calc, Statistics, Physics: (Mechanics, E&M, Fluids, Waves, Thermodynamics, Optics), Signals and Systems, Circuit Components, Probability and Stats, Stochastic Processes, Linear Control Systems, Linear Electronic Circuits, Linear Systems, Linear & Nonlinear Optimization, Digital Comm Theory, Data Networks, Signal Processing, Deep Learning, Machine Learning, Radar Theory and Lab, Digital Communication Lab, Image Processing, Electromagnetism

AWARDS AND HONORS

- **UCSD Jacobs Engineering Scholarship**: Merit based full-ride scholarship, awarded based on academics, leadership, and innovative potential.
- x8 UCSD Provost Honors
- 1st Place Award at 2023 GSDSEF (Greater San Diego Science & Eng. Fair), CSEF Qualifier (CA Science & Eng. Fair)
- AIME Qualifier (American Invitational Math Competition; for high scoring AMC 10/12 participants)
- BMES High School Poster Conference Honorable Mention
- Dean's List Award Semi-finalist at 2021 FIRST Robotics Competition

PUBLICATIONS

- B. Chen, **J. Chen**, W. Zhao, K. Zhang, X. Zhang. "**MetaVenom: Field Programmable Passive Metasurfaces,**" ACM MobiCom 2026.
- W. Zhao, B. Chen, K. Zheng, X. Chen, W. Zhang, X. Zhang. "**FlowForm: Scalable Passive Metasurface Network for mmWave Coverage Expansion**," ACM SIGCOMM 2026. (**in Acknowledgements**)
- Y. Li, S. Wang, Y. Fu, **J. Chen**, et. al. "**UltraPoser: Pushing the Limits of IMU-based Full-Body Pose Estimation with Ultrasound Sensing on Consumer Wearables,**" In ACM Symposium on User Interface Software and Technology (UIST), Sep. 2025

RESEARCH/WORK EXPERIENCE

Yale Department of Electrical & Computer Engineering *Jun. 2026 - Present*

Research Intern under Prof. Tara Boroushaki

mmWaveX: Cross Polarization Radars for Real Time Scene Reconstruction

- Using transformer based neural scene reconstruction for 2 cross polarization FMCW radar to derive environment 3d occupancy map.
- Implemented hardware trigger radar sync on TI AWR1843 and IWR1443 using programmed Arduino external clock.
- Engineered data collection setup for mmWave radar perception with depth camera ground truth and facilitated simultaneous frame firing through strict hardware trigger and ROS supervision.
- Created Flask based dashboard website that facilitated collection debugging, collection, and management.
- Set up and storage server with strict cybersecurity protocols for storing collected data and uploading to compute clusters.
- Implementing early sensor fusion transformer architecture for flexible NLOS capable scene reconstruction.

UCSD (UC San Diego) Center for Wireless Communications *Oct. 2024 - Present*

Research Intern under Prof. Xinyu Zhang

mmGrasp: Adaptive Sparse Aperture Tomography Sensing for Material and Geometry Reconstruction

- Researching radar perception for robotics and grasping – enabling flexible material aware perception for NLOS and difficult grasping scenarios.
- Hacked Infineon BGT60TR13C FMCW radars through custom SPI trigger commands facilitated by Xilinx Artix 7 FPGA to enable multistatic RSS signal reception for sparse array tomography setup.
- Engineering CNN based classification algorithm for material perception based on monostatic received signal and multistatic RSS
- Designed 2d rotational tomography data collection setup with ST2315 servos for FMCW radar angular sweeping.
- Implementing filtered backprojection based object reconstruction from tomographic RSS data and comparison to 2d FDTD simulation for digital twin reconstruction.

MetaVenom: mmWave Quasi-Passive Metasurfaces for Network Enhancement and JCAS

- Researching and developing new **quasi dynamic passive metasurfaces for mmWaves (60Ghz)** to increase signal coverage that allows for passive reconfigurability using ferrofluid. Our methods reduce are much cheaper than fully active metasurfaces and overcome the weaknesses of passive surfaces not being reconfigurable.

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- Implemented and tested joint optimization algorithm using **differential evolution genetic algorithm and gradient fine tuning** that outputs metasurface design and ferrofluid configuration based on incident signal direction and desired output beam patterns. Resulting evaluated beam patterns are close to simulation with around clear **10-15dB peak RSS increase at designed angles at 5m**.
- Presented preliminary results at UCSD Summer Research Conference. Published in **Mobicom 2026**.

FlowForm: mmWave Network Coverage Enhancement using Networked Passive Metasurfaces

- Developing passive networked metasurface/reflectarray design and placement optimization alg. for mmWave (60Ghz).
- Tested and validated hierarchical placement and design optimization algorithm using gradient descent and differentiable ray tracing on custom FPGA controlled 60Ghz software defined radio (SDR). Algorithm optimized parameters given environmental constraints, designing for robustness including channel dynamics, interference, multiple users, and nonlinearities, demonstrating a **10+% received signal strength (RSS) + coverage increase from baselines.**
- Presented preliminary results with Poster at DevCom NC4 Technical Review @ UCR. Published in **SIGCOMM 2026**.

UltraPoser: IMU Based Human Pose Estimation Enhanced by Acoustic Sensing

- Researched and implemented multimodal **pose estimation model using IMU and Acoustic Sensing** on consumer wearables: phone, watch, and earbud. Used Acoustic Data to cross validate IMU sensor data, through doppler frequency shift and CIR multipath propagation.
- Designed data collection procedure & preprocessed data for multimodal fusion transformer-based model. Model uses GCN, TCN, Transformers, Cross attention, and multi-modal fusion, achieving **~30% decrease in joint position error** vs IMU only methods.
- Presented Results in 2025 UCSD Undergraduate Research Conference. Published in 2025 ACM UIST.

US Navy: Naval Surface Warfare Center, Philadelphia Division *Jun. 2024 – Sep. 2024*

Research Intern

- Acquired knowledge and expertise in Fluid Dynamics, Reinforcement Learning, and Design Optimization: Implemented a standalone application using MATLAB and Simulink (Signal Processing, Deep Learning, Reinforcement Learning Toolboxes) that used a **novel Machine Learning** (Reinforcement Learning) and Computational Fluid Dynamics based approach for **axial fan blade design optimization**.
- Employed a customized **Actor-Critic Deep Deterministic Policy Gradient Agent and Finite Difference Method solution of the Navier Stokes Equation** to simulate and optimize fan designs based off parameters such as fluid speed, viscosity, etc.

STEM EXTRACURRICULARS

Robotics Club *Oct. 2019 – Jun. 2023*

Vice President and Head of Design for Team 3647

- Constructed an **Odometry robot positioning system using magnetic encoders and IMU sensors**, enabling precise robot navigation & location tracking. **Reduced positional errors from ~1 meter to 10 cm**, enabling significantly higher task completion rates.
- Developed knowledge in robot control systems, implementing a PID control algorithm for subsystems such as robot arm in order to enable **precise movement with rotational accuracy of about 1 degrees**. Led to robot winning the Idaho Regional Competition.

SERVICES AND CLUB ACTIVIES

- Finance Vice President @ Cornerstone Community Consultants *Oct. 2025 – Present*
- Funding Manager @ Association of Comp. Machinery (ACM), UCSD Student Branch *Nov. 2023 – Jun. 2025*
- Sponsorship Manager @ Eta Kappa Nu (HKN), UCSD Student Branch *July. 2024 – Jun. 2025*
- Membership Chair for Scholar Welcome Society @ UCSD Scholar Society *Oct. 2023 – Jun. 2024*
- CAD and Programming Tutor *Oct. 2022 – Jun. 2023*

ADDITIONAL SKILLS

- Computer Aided Design (AutoCAD, OnShape, AutoDesk Fusion 360, Solidworks), CAM (Comp. Aided Manufacturing)
- Fabrication (3D Printing, Milling, Lathing, CNCing)
- Programming (Python, MATLAB, Simulink, Javascript, C, C++, Assembly, MIPS)
- AI and Machine Learning (OpenCV, Pytorch, SciPy, Numpy, Tensorflow, Pandas, etc)
- Robot Controls with PID, RL, Model Predictive Control
- Finite Element Analysis, CFD (Fusion 360, Ansys HFSS)
- KiCAD, Eagle, FPGAs, PSpice, LTSpice, Verilog